

## SYNOPSIS

At 0218 on 6 March 2011, the UK registered container ship *Cosco Hong Kong* collided with the China registered fish transportation vessel *Zhe Ling Yu Yun 135*. The accident occurred in international waters off the coast of Zhejiang Province, China. *Zhe Ling Yu Yun 135* sank almost immediately, with the loss of 11 lives. *Cosco Hong Kong* was not damaged.

On impact, *Cosco Hong Kong*'s officer of the watch (OOW) felt an unusual vibration and immediately put the engine telegraph to stop. He did not see *Zhe Ling Yu Yun 135* immediately before or after the collision and he was unaware of what the container ship had struck. *Cosco Hong Kong* remained drifting in the vicinity for over 1 hour while the master tried to establish what had happened. In the absence of any evidence that a collision had occurred, *Cosco Hong Kong* then resumed passage to Yangshan, China.

The Taizhou Maritime Rescue Co-ordination Centre was notified that *Zhe Ling Yu Yun 135* was missing at 2021, 18 hours after the collision. An air and sea search failed to find any trace of the vessel or her crew. The wreck of *Zhe Ling Yu Yun 135* was eventually located on 17 March 2011 close to the position of the collision.

Factors that led to the collision included:

- The performance of *Cosco Hong Kong*'s OOW fell well short of expected standards. He did not correctly apply the collision regulations or follow onboard instructions.
- *Zhe Ling Yu Yun 135* and *Cosco Hong Kong* turned towards each other at about the same time when only about 1.5nm apart and with a closing speed of over 20kts.
- *Cosco Hong Kong*'s OOW was the sole lookout. In darkness, in adverse weather conditions, and among large concentrations of fishing vessels, he did not see that *Zhe Ling Yu Yun 135* was approaching.
- Even though many fishing vessels were concentrated in the area, *Cosco Hong Kong*'s master or OOW did not consider amending the voyage plan, reducing speed, or enhancing the bridge manning.

Recommendations have been made to the China Ocean Shipping Company (COSCO) and COSCO Maritime UK Ltd (Cosmar) intended to improve bridge watchkeeping standards on all vessels managed by Cosmar and other COSCO subsidiaries. A recommendation has also been made to the Ministry of Agriculture of the People's Republic of China aimed at increasing the survivability of Chinese fishermen following marine casualties.

External & internal  
environment

The visibility was 3nm in rain, the sea state was moderate, and the wind was from the north-north-west force 5.

Persons on board

23

11

Photograph courtesy of [www.fotoflite.com](http://www.fotoflite.com)



*Cosco Hong Kong*



*Zhe Ling Yu Yun 135*

## 1.2 NARRATIVE

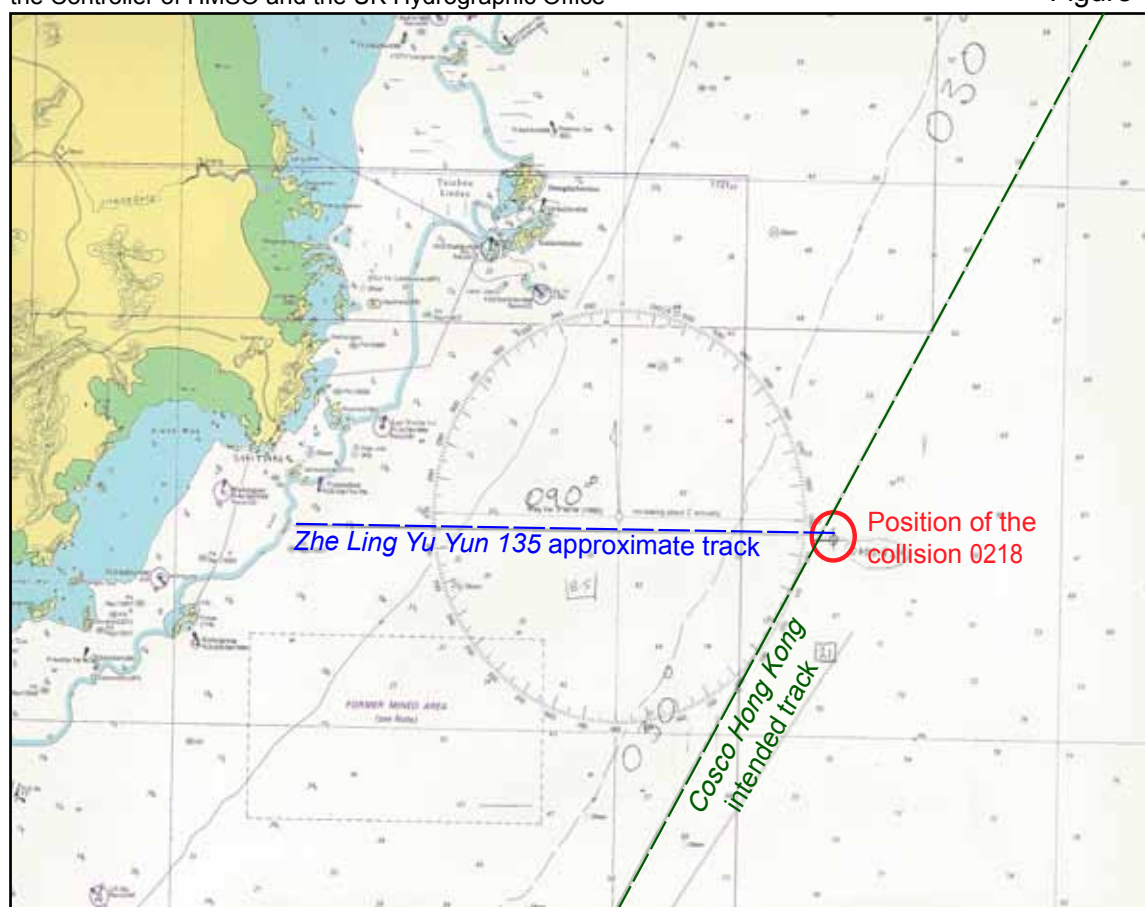
### 1.2.1 The collision

At 2209 on 5 March 2011, the fish transportation vessel *Zhe Ling Yu Yun 135* sailed from Shi Tang, Zhejiang Province, China (**Figure 1**). The vessel had 11 crew and was bound for fishing grounds 150nm to the east at a speed of about 8.5kts. *Zhe Ling Yu Yun 135*'s identity, position, course over the ground (COG) and speed over the ground (SOG) were being transmitted via her Automatic Identification System (AIS).

*Cosco Hong Kong* was on passage from Hong Kong to Shanghai, China using autopilot-controlled headings to follow a planned track of 030° (**Figure 1**). The vessel's speed was 21kts.

Reproduced from Admiralty Chart BA 1759 by permission of the Controller of HMSO and the UK Hydrographic Office

Figure 1



Extract of chart BA 1759

At about 2345, *Cosco Hong Kong*'s master completed writing his night orders (**Annex A**) and then went to bed in his cabin. At 2350, the second officer relieved the third officer as the officer of the watch (OOW). The second officer was accompanied on the bridge by an able seaman (AB) who was the nominated lookout. The visibility was good, the sea state was moderate, and the wind was from the north-north-west force 5.

At about 0150 on 6 March 2011, the OOW gave the AB permission to leave the bridge to conduct a fire patrol. This was a usual practice during which the AB also cooked noodles in the galley. By now, heavy rain had reduced the visibility to about 3nm. It had also adversely affected the quality of the radar picture as shown on the bridge radar displays.

Between 0153 and 0157, the second officer gradually altered the autopilot-controlled heading from 030° to 055° in order to keep clear of a group of near-stationary fishing vessels directly ahead of *Cosco Hong Kong* at a range of between 5nm and 6nm (**Figures 2 and 3**). *SITC Pyeongtaek*, a 9530t container ship making good a COG of 009° and an SOG of 15kts, was 5nm on *Cosco Hong Kong*'s starboard side. *Cosco Hong Kong* was overtaking *SITC Pyeongtaek*.

By about 0205, the distance between *Cosco Hong Kong* and *SITC Pyeongtaek* had reduced to approximately 2nm. *SITC Pyeongtaek*'s OOW was concerned that *Cosco Hong Kong* would pass only 3 cables ahead of his vessel, and he called *Cosco Hong Kong* in Mandarin via Very High Frequency (VHF) radio channel 16. No response was heard, so *SITC Pyeongtaek*'s OOW altered his vessel's heading to starboard to about 025° in order to increase the closest point of approach (CPA) of the overtaking vessel.

Between 0205 and 0210, *Cosco Hong Kong*'s OOW adjusted the autopilot-controlled heading to about 040°, and again at 0210 to about 048°. During this period, he saw a second group of near-stationary targets on the radar approximately 6nm ahead (**Figure 4**). The OOW also saw an east-moving radar target, later identified as *Zhe Ling Yu Yun 135*, to the north of the closest group of fishing vessels. The OOW estimated that *Zhe Ling Yu Yun 135* was crossing from the port bow. However, as the OOW was not monitoring the AIS receiver or the 'X' band radar which was able to display AIS targets, he was not aware that this vessel was *Zhe Ling Yu Yun 135*. *Cosco Hong Kong* was now to the east of her intended track.

At 0214, *Cosco Hong Kong* was passing ahead of *SITC Pyeongtaek* at a distance of 6 cables; she was also clearing the group of fishing vessels on her port side (**Figure 5**). However, the second officer remained concerned by the movement of *Zhe Ling Yu Yun 135*, which had closed to within 2nm. The OOW sounded one blast lasting approximately 3 seconds on the ship's forward whistle but *Zhe Ling Yu Yun 135* maintained her course and speed.

The OOW decided to leave the fish transporter on his vessel's starboard side by making a bold alteration of course to port to a heading of 330°. He considered that this action would not only eliminate the risk of collision with *Zhe Ling Yu Yun 135*, but it would also keep *Cosco Hong Kong* clear of the numerous fishing vessels ahead and move the container ship back towards her intended track.

The second officer checked the 'S' band radar display sited on the starboard side of the bridge (**Figure 6**) to confirm that the intended heading of about 330° was clear of other vessels. He then moved to the port bridge wing and looked over the port bow and the port beam to ensure that it was safe to alter course. The OOW then returned to the centreline console and began to adjust the vessel's heading to port using the joystick control on the vessel's track pilot system.

At 0215, *Cosco Hong Kong* began turning to port. *SITC Pyeongtaek*'s OOW could now see a white light from *Zhe Ling Yu Yun 135*, which was directly ahead of his vessel at a distance of about 2nm. *SITC Pyeongtaek*'s OOW had also not identified *Zhe Ling Yu Yun 135* by AIS; and he no longer saw the white light from the fish transporter after the *Cosco Hong Kong* again crossed in front of his vessel's bow.

Between 0215 and 0216, *Zhe Ling Yu Yun 135* altered course to starboard to make good a COG of 220°, towards the path of *Cosco Hong Kong* (**Figure 7**). *Cosco Hong Kong*'s OOW also moved to the port wing to check that the container ship was turning into clear water and then returned to the 'S' Band radar display. He did not see *Zhe Ling Yu Yun 135* alter course. The two vessels began to close rapidly and at 0218:08, *Cosco Hong Kong* and *Zhe Ling Yu Yun 135* collided.

### 1.2.2 Post-collision

On impact, *Cosco Hong Kong*'s OOW felt a sudden and unusual vibration from the forward part of the vessel. The OOW was unaware of what had caused the vibration and he quickly put the engine telegraph to stop. He did not know that his vessel had collided with the fish transporter. The vibration was also felt by the chief officer, who was awake in his cabin two decks below the bridge. He immediately telephoned the OOW and asked what had happened. The OOW replied that '*the ship might have collided with another vessel*'.

The OOW then telephoned the master, who was asleep in his cabin, and informed him that he had felt an unusual vibration and that *Cosco Hong Kong* might have collided with another vessel; he also asked the master to come to the bridge. The second officer then telephoned the crew mess and galley and recalled the AB back to the bridge. He also switched on the deck lights so that he could see the sea surface close by. The OOW went onto the port and starboard bridge wings but he did not see anything unusual.

The master arrived on the bridge at about 0220 to find *Cosco Hong Kong* on a heading of 300° and reducing speed. The master quickly saw *SITC Pyeongtaek* close on the port quarter and passing astern. He also noticed the group of fishing vessels on the port side at a distance of 0.8nm, and another group of fishing vessels 45° off the starboard bow at a distance of 2.1nm. By 0224, *Cosco Hong Kong* was heading 309° at a speed of 9kts.

The master asked the OOW what had happened, why the cargo ship astern was so close, and why the deck lights were switched on. The second officer did not answer the master's questions. Instead, he went to the starboard bridge wing to monitor *SITC Pyeongtaek* passing astern. When the OOW returned inside the bridge, the master again asked him what had happened. The second officer informed him that a small seagoing vessel had approached *Cosco Hong Kong* from the port side but she had not taken any action after he had sounded the whistle. The second officer was turning the container ship to port when he had felt an unusual vibration. The second officer confirmed to the master that he did not see anything ahead of the ship, either by radar or visually, during the course alteration.

The master was concerned that *Cosco Hong Kong* might have grounded or collided with something. He instructed the second officer to log the vessel's position and then checked that the vessel was clear of navigational dangers. He also arranged for the chief officer, the bosun and the carpenter to check for damage on the upper

*In areas of high traffic density, in conditions of restricted visibility and in all hazardous navigational situations ensure the vessel is in hand steering.*

*Station a person to steer the vessel and to put the steering into manual control in good time to allow any potentially hazardous situation to be dealt with in a safe manner. Officers are further reminded that when the vessel is in automatic steering it is highly dangerous to allow a situation to develop to the point where the OOW is without assistance and has to break the continuity of the look-out in order to take emergency action. [sic]*

## **1.9 REGULATION**

### **1.9.1 COLREGS**

Rules 2, 5, 6, 8, 13, 15, 16, 17 and 34 of the COLREGS are at **Annex F**.

### **1.9.2 United Nations Convention on the Law of the Sea (UNCLOS)**

The UNCLOS requires the master of any ship involved in a collision to stand by and offer assistance to the other vessel:

*Article 98 - Duty to render assistance*

*1. Every State shall require the master of a ship flying its flag, in so far as he can do so without serious danger to the ship, the crew or the passengers:*

*(a) to render assistance to any person found at sea in danger of being lost;*

*(b) to proceed with all possible speed to the rescue of persons in distress, if informed of their need of assistance, in so far as such action may reasonably be expected of him;*

*(c) after a collision, to render assistance to the other ship, its crew and its passengers and, where possible, to inform the other ship of the name of his own ship, its port of registry and the nearest port at which it will call.*

### **1.10 CHINESE FISHING VESSEL REGULATION**

Chinese fishing industry vessels, including fish transporters, are administered and regulated by the Fisheries Department of the Chinese Ministry of Agriculture. Over 280,000 fishing vessels are registered in China, but it is estimated that up to 1 million Chinese fishing vessels operate along the Chinese coast.

The cost of fitting the AIS on board *Zhe Ling Yu Yun 135* was jointly subsidised by the Zhejiang Provincial Government and the Chinese Ministry of Agriculture as part of an initiative to improve fishing vessel safety. Over 16000 fishing vessels in Zhejiang Province have been fitted with Class B AIS through this scheme.

The Chinese Ministry of Agriculture requires registered fishing vessels to carry liferafts secured by hydrostatic release units (HRUs), but it has found that many fishing vessel skippers also tie down the liferafts in order to prevent them from being lost overboard in rough weather.

## 1.11 PREVIOUS ACCIDENTS

### 1.11.1 Cosmar vessels

In 2003, *Cosco Rotterdam* collided with a Taiwanese fishing vessel resulting in four fatalities among the fishing vessel's crew. No details of this accident have been found.

In 2004, *Cosco Hamburg* collided with *PO Nedlloyd Finland* on the River Elbe, Germany. Both vessels were being manoeuvred as advised by embarked river pilots. An investigation was conducted by the Bundestelle für Seefalluntersuchung (BSU) (German Federal Bureau of Marine Casualty Investigation) which highlighted that the collision occurred as a result of interaction as *Cosco Hamburg* overtook *PO Nedlloyd Finland*, which caused the latter vessel to lose control of her steering.

In 2009, the MAIB conducted a preliminary examination into the grounding of *Cosco Hong Kong* off Hong Kong, following which the Chief Inspector of Marine Accidents advised Cosmar:

- It is apparent that the OOW became over-loaded when trying to avoid a large number of small fishing vessels in the Dadanwei Shuidao channel and forgot about the presence of the Lixin Pai Reef. This could have been avoided had, when the passage was planned, proper consideration been given to the risks of encountering high traffic densities and the proximity of hidden dangers in this area.
- A critical review of the plan might have prompted a number of precautions to be taken when transiting the channel such as; proceeding at a slower speed to allow the OOW more time to meet the potential demands of collision avoidance as well as navigational safety, and the provision of additional support to enable the ship's position to be quickly and accurately monitored by several methods, including radar parallel indexing and the use of the vessel's electronic chart system.
- Therefore, in order to try and prevent similar accidents occurring in the future, I strongly advise that you take measures to ensure that all passage plans are critically reviewed by masters to identify the areas in which routine bridge manning potentially requires enhancement, and where the speed of transit requires careful consideration. I also advise that you stress to all bridge watchkeeping officers the importance of calling the master when in doubt or when having difficulty in keeping a vessel safe.

### 1.11.2 Chinese registered fishing vessels

The MSA Annual Report 2010 details Chinese fishing vessel casualties in 2009 and 2010 as:

| Year/Type | Foundering | Fatality |
|-----------|------------|----------|
| 2009      | 42         | 109      |
| 2010      | 38         | 146      |

In 2009, there were 11 collisions involving fishing vessels that resulted in three or more fatalities. The report also identified that UK registered vessels had been involved in five accidents or incidents in Chinese waters since 2006.

### 1.11.3 Recent similar accidents

There have been four recent accidents between merchant vessels and UK registered fishing vessels that have been investigated by the MAIB and have similarities with the collision between *Cosco Hong Kong* and *Zhe Ling Yu Yun 135*.

#### *Alam Pintar/Etoiles des Ondes* ([MAIB Report No 11/2010](#))

On 20 December 2009, the Singapore registered bulk carrier *Alam Pintar*, collided with the fishing vessel *Etoiles des Ondes* in the English Channel. As a result of the collision, *Etoiles des Ondes* sank and one of her crew was lost. The remaining crew were rescued from the vessel's liferaft by a passing ferry. The fishing vessel was carrying an EPIRB, which activated soon after she foundered.

*Alam Pintar's* OOW (fourth officer) was an inexperienced officer and, although he had seen *Etoiles des Ondes* and realised that there was a risk of collision, his initial alterations of course were rendered ineffective when the fishing vessel also changed course to start shooting her pots. Although *Alam Pintar's* OOW ordered the helm hard-a-starboard, his action was not effective in avoiding the collision. Both *Alam Pintar's* master and OOW were aware of the collision but did not stop the vessel, report the incident, or attempt to check if *Etoiles des Ondes's* crew were safe.

#### *Scottish Viking/Homeland* ([MAIB Report No 4/2010](#))

On 5 August 2010, the Italian registered ro-ro passenger ferry *Scottish Viking* collided with the fishing vessel *Homeland* about 4 miles off St Abbs Head, England. As a result of the collision, the fishing vessel sank and one of her crew was lost. As the fishing vessel sank, her liferaft was automatically released and it inflated on reaching the surface.

Among the factors that led to the collision identified in the MAIB investigation report was that *Scottish Viking's* OOW (second officer) did not: determine at an early stage if there was a risk of collision with *Homeland*; sufficiently monitor or plot *Homeland's* track; and, once a risk of collision was deemed to exist, take sufficient action to avoid a collision.

Following the collision, *Scottish Viking's* manager took action aimed at improving the performance of the company's bridge teams, including reiterating the importance of following the company's navigational procedures, carrying out unscheduled navigational audits at sea, and randomly scrutinising VDR data.

#### *Boxford/Admiral Blake* ([MAIB report 17/2011](#))

On 11 February 2011, the Marshall Islands registered container ship *Boxford* collided with the fishing vessel *Admiral Blake* in the English Channel. On impact, two of *Admiral Blake's* deckhands were thrown overboard, but both were safely recovered; one of them was rescued by the crew of *Boxford's* rescue boat, which had been launched to assist. *Admiral Blake* was badly damaged and had to be towed to Plymouth, England.

The MAIB investigation report identified that *Boxford*'s bridge team was unaware of the presence of *Admiral Blake* until shortly before the collision. Neither the radar nor the visual lookout was fully effective and the vessel's master inaccurately assessed the fishing vessel's proximity and movement. The resulting manoeuvres taken were hazardous and resulted in the collision. A recommendation was made to the vessel's manager aimed at improving the standard of lookout and bridge watchkeeping across its fleet.

*Philipp/Lynn Marie* ([MAIB Report 20/2011](#))

On 9 April 2011, the Gibraltar registered container feeder vessel *Philipp* collided with the scallop dredger *Lynn Marie* in the Irish Sea, 6nm south of the Isle of Man. There were no injuries, but *Lynn Marie* was badly damaged and had to be towed to Port St Mary, Isle of Man.

The MAIB investigation report identified that *Philipp*'s OOW (chief officer) had not properly assessed the risk of collision with *Lynn Marie* and that his actions taken to try and avoid the fishing vessel were unnecessary and contrary to the requirements of the COLREGS. A recommendation was made to *Philipp*'s manager aimed at raising the competency of its bridge watchkeeping officers and ensuring that the navigation and anti-collision aids fitted on board its vessels were used effectively.

## SECTION 2 - ANALYSIS

### 2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

### 2.2 RISK OF COLLISION

It is evident from the OOW's adjustment of *Cosco Hong Kong's* heading to 055° between 0153 and 0157 that he was aware of, and was taking action to avoid, the group of fishing vessels between 5nm and 6nm directly ahead of the bow (**Figures 2 and 3**).

Before the alteration to 055°, *Zhe Ling Yu Yun 135* was crossing ahead of *Cosco Hong Kong* at a distance of 7nm and would have passed down the container ship's starboard side with a CPA of over 2nm. Similarly, *SITC Pyeongtaek* had almost been overtaken by *Cosco Hong Kong* and was also passing well clear. Following the alteration to 055°, *Cosco Hong Kong's* CPA with the fish transporter was reduced to approximately 2 cables on *Cosco Hong Kong's* port quarter, and the container ship was predicted to pass only 3 cables ahead of *SITC Pyeongtaek*. In effect, in taking action to avoid the drifting fishing vessels, *Cosco Hong Kong's* OOW had caused a risk of collision with two vessels that had previously been passing clear.

The adjustments to *Cosco Hong Kong's* heading between 0205 and 0210 would have led to some increase of the CPA with *SITC Pyeongtaek*. However, most of the increase in CPA from 3 cables to 6 cables can be attributed to the earlier action taken by the overtaken vessel's OOW. However, the adjustment to *Cosco Hong Kong's* heading had little effect on her CPA with *Zhe Ling Yu Yun 135*, which remained steady at about 2 cables. Although the container ship was now approaching *Zhe Ling Yu Yun 135* from abaft the fish transporter's starboard beam, the fish transporter was still crossing ahead. In accordance with Rule 15 of the COLREGS (**Annex F**), *Zhe Ling Yu Yun 135* was the give way vessel and *Cosco Hong Kong* was the stand on vessel. The onus to take avoiding action rested with *Zhe Ling Yu Yun 135*.

### 2.3 DECISION-MAKING

#### 2.3.1 The use of a sound signal

Having assessed that there was a risk of collision between *Cosco Hong Kong* and *Zhe Ling Yu Yun 135*, the second officer's expectation that the fish transporter would take avoiding action was justified. When this did not happen by the time the vessels had closed to within 2nm at 0214, his decision to alert the crew on board *Zhe Ling Yu Yun 135*, by sounding the ship's whistle, was appropriate.

However, Rule 34 of the COLREGS (**Annex F**) provides clear instructions for the use of manoeuvring and warning signals when vessels are in sight of one another. In this case, the sound signal that should have been used by *Cosco Hong Kong's* OOW to indicate his concern over the actions and intentions of the crew on board *Zhe Ling Yu Yun 135* was at least five short blasts.

The OOW's use of one blast of approximately 3 seconds, which is not prescribed in the COLREGS, was therefore incorrect. It could easily have been mistaken on board *Zhe Ling Yu Yun 135* for one short blast, indicating that *Cosco Hong Kong* was altering course to starboard. Nonetheless, it is possible that the sound signal did alert *Zhe Ling Yu Yun 135*'s crew to the approach of the container ship and prompted her wheelhouse watchkeeper to alter the fish transporter's heading boldly to 220°.

### 2.3.2 Alteration to port

Rule 17 of the COLREGS (**Annex F**) allows a stand-on vessel to take action by her manoeuvre alone as soon as it becomes apparent that the vessel required to keep out of the way is not taking appropriate action, Rule 17 (c) states:

*A power-driven vessel which takes action in a crossing situation...shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.*

In this case, there were no navigational constraints or other vessels that prevented *Cosco Hong Kong*'s OOW from altering course to starboard at any time, or from reducing speed once the container ship was clear of *SITC Pyeongtaek*. However, the second officer decided to alter the vessel's heading to port. Although this decision might have been influenced by the advantages of the alteration taking *Cosco Hong Kong* back towards her intended track and leaving the next group of fishing vessels clear on the starboard side, it was contrary to the requirements of the COLREGS. Consequently, *Cosco Hong Kong*'s alteration to port would not have been expected by the crew on board the other vessels in the vicinity, including *Zhe Ling Yu Yun 135* and *SITC Pyeongtaek*.

## 2.4 MONITORING OF AVOIDING ACTION

### 2.4.1 Cosco Hong Kong

**Figure 7** shows that when *Cosco Hong Kong*'s heading began to alter to port at 0215, *Zhe Ling Yu Yun 135*'s heading was altered to starboard at almost the same time. However, although *Zhe Ling Yu Yun 135* eventually turned over 120° to a new heading of about 220°, neither this bold alteration nor the fish transporter's approach were seen by *Cosco Hong Kong*'s OOW, despite the fish transporter's lights being visible to the more distant *SITC Pyeongtaek*'s OOW.

Rule 8 of the COLREGS states:

*Action taken to avoid collision with another vessel shall be such as to result in passing at a safe distance. The effectiveness of the action shall be carefully checked until the other vessel is finally past and clear*

In this case, the rule was clearly not followed by the OOW. This was probably due to a combination of several factors.

First, the OOW had allowed the AB to leave the bridge, which left the OOW as the sole lookout.

Second, in using the joystick control on the vessel's track pilot system to alter *Cosco Hong Kong's* heading, the OOW had overridden the autopilot and was in hand-steering. Consequently, he was also the helmsman, and his effectiveness as a lookout was reduced.

Third, *Zhe Ling Yun 135's* radar target might have been lost within the clutter near the origin of the S-band radar display the OOW was monitoring, particularly as the 6nm range scale was selected and the quality of the display was affected by rain. The OOW was also not monitoring AIS information.

Fourth, it is highly likely that, during the 3 minutes from the start of the course alteration until the collision, the OOW focused on ensuring that *Cosco Hong Kong* was turning into clear water. It is also possible that the OOW considered that as he was making such a bold alteration, *Zhe Ling Yu Yun 135* was no longer a problem and did not need watching.

Finally, *Zhe Ling Yu Yun 135* would not have been visible from *Cosco Hong Kong's* bridge when the fish transporter was within 567m of the container vessel's bow because of the shadow zone ahead of the vessel due to the deck cargo carried. Given a closing speed of about 20kts at that point, the fish transporter would not have been visible to the OOW for approximately the 50 seconds immediately before the collision. Although the shadow zone of *Cosco Hong Kong* exceeded the maximum of 500m as prescribed in the International Convention for the Safety of Life at Sea (SOLAS), the excess 67m equated to approximately 6 seconds, during which the fish transporter could not be seen, and is highly unlikely to have been contributory to the accident in this case.

#### **2.4.2 *Zhe Ling Yu Yun 135***

It must be assumed that *Zhe Ling Yu Yun 135's* alteration of heading towards 220° between 0215 and 0216 was initiated by her wheelhouse watchkeeper in order to avoid *Cosco Hong Kong*. **Figure 5** shows that the fish transporter was passing ahead of *SITC Pyeongtaek* and there were no other vessels transmitting on AIS that were of concern. It will never be known if the alteration was prompted by the watchkeeper hearing the single blast on the container ship's whistle.

When *Zhe Ling Yu Yun 135* began to alter course to starboard, it is estimated that she was about 30° on *Cosco Hong Kong's* port bow. On steadying on a heading of about 220° at 0217:09, the fish transporter was now only about 6° on the container ship's port bow. It is not known if the wheelhouse watchkeeper intended to make such a large alteration or whether he continued to turn on seeing that *Cosco Hong Kong* was altering to port. Nonetheless, the container ship's aspect was likely to have been confusing to the wheelhouse watchkeeper, and the fish transporter was now only 4 cables from the container ship's bow and closing rapidly. The wheelhouse watchkeeper therefore had only about 1 minute in which to re-assess the situation and take avoiding action. Given that the watchkeeper might not have realised that *Cosco Hong Kong* remained under helm to port, this was an extremely difficult situation and it is evident from the collision that whatever further avoiding action he might have attempted it was not successful.

## 2.5 RELEASE OF THE LOOKOUT

In view of the fact that *Cosco Hong Kong's* OOW did not notice that *Zhe Ling Yu Yun 135* altered course to 220°, or see her as the vessels continued to close, the release of the AB lookout from the bridge was potentially pivotal to this accident. Had the lookout remained on the bridge, it is possible that he would have seen the approach of *Zhe Ling Yu Yun 135* in sufficient time for avoiding action to be taken. If necessary, the AB would then have been available to take the helm. Even if a collision could still not have been avoided, the presence of the AB would have at least enabled the OOW to monitor the fish transporter immediately after the collision and ensure that the alarm was quickly raised.

Many bridge watchkeepers interviewed by the MAIB in recent years have considered the employment of deck ratings on the bridge as an additional lookout to have little value, even during darkness or in busy shipping areas. In many cases where they are used, their presence is seen as a token gesture aimed at meeting regulatory requirements. Others simply prefer to keep their watches alone.

In this case, although the release of the lookout during the midnight to 0400 bridge watch was contrary to the vessel's onboard instructions for the safety of navigation (**Annex E**), and the guidance provided in MGN 315 (Paragraph 1.8), it had become a routine practice. In allowing the lookout to leave the bridge at a critical time, when the vessel was on a night passage through dense shipping in adverse weather conditions, the second officer appears not to have appreciated the value of the lookout or the dangers developing ahead.

## 2.6 EXECUTION OF THE VOYAGE PLAN

In view of the size of China's fishing fleet, it is not surprising that dense concentrations of fishing vessels are frequently encountered along its coast. While the fishing is seasonal, peaking during the summer months, similar numbers of fishing vessels to those shown in **Figures 2 to 5** can be expected to be encountered in this area, and other coastal areas of China, at any time of the year.

Like many container ships on a liner service, *Cosco Hong Kong* was operating on fixed routes and her voyage plan between Hong Kong and Shanghai, including the track along the coast of Zhejiang Province, had probably been used without difficulty on numerous previous occasions. Nevertheless, although the master indicated in his night orders (**Annex A**) that he was aware that large numbers of fishing vessels might be encountered, these did not specify any precautionary measures that could be observed in such circumstances. Consequently, when the second officer detected the first group of fishing vessels ahead of the bow shortly before 0200, he continued to try and follow the intended track at the planned speed of 21kts. This action was clearly inappropriate and unnecessary.

It is evident from **Figures 2 to 5** that *Cosco Hong Kong* could have avoided most of the fishing vessels in the vicinity had her voyage plan been adjusted further to the east, similar to the tracks of other vessels including *World Trader*, *Dong-Z Calypso* and *Odigitria*, which were also transiting the area. Although this would have increased the length of her passage, the overall effect would have been negligible, particularly as the vessel would probably have been able to safely continue at 21kts.

Alternatively, a reduction in speed would have provided *Cosco Hong Kong's* OOW with more time to assess any risk of collision and to take appropriate avoiding action. Given the reduced visibility in rain showers, the concentrations of fishing vessels, and the effect of the sea state and rain on radar detection, such action was also warranted by Rule 6 of the COLREGS (**Annex F**).

The relatively high speed of container ships in areas of high traffic density and/or restricted visibility has been identified as a contributory factor in many marine casualties; the grounding of *Maersk Kendal* (MAIB report 2/2010) is a recent example. Although *Cosco Hong Kong's* passage speed of 21kts was lower than the vessel's service speed, this reduction had been made to save fuel, rather than to aid safe navigation. Moreover, the passage speed was still considerably faster than other vessels in the area.

Finally, although the vessel's instructions for the safety of navigation (**Annex E**) stated that the master:

*Should be present on the bridge supervising Officers' work and take over the navigating himself when the ship is in an area of heavy shipping or where fishing boats are dense [sic]*

this did not happen in this case. Had the master been on the bridge, he could have enhanced the effectiveness of the bridge team through his knowledge and experience. He would also have been best placed to assess and mitigate the risks posed by the concentrations of fishing vessels by amending the vessel's route or by reducing speed. It is highly unlikely that the master would have allowed the lookout to leave the bridge.

It is disappointing that several of the lessons to be learned from *Cosco Hong Kong's* grounding in 2009 (Paragraph 1.11.1), particularly the need to critically review passage plans in order to identify the areas in which routine bridge manning requires enhancement, and where the speed of transit requires careful consideration, were again not taken into account on this occasion. If further similar accidents are to be avoided in the future, it is important that masters and deck officers are made aware of the need to continually assess the risks to a vessel's safe navigation during a passage and, where necessary, implement appropriate mitigating measures.

## **2.7 ACTION TAKEN FOLLOWING THE COLLISION**

Immediately after feeling the vibration caused by the impact of *Zhe Ling Yu Yun 135*, *Cosco Hong Kong's* OOW was undoubtedly concerned that his vessel had struck an object in the water. However, because he had not seen the fish transporter during the turn or after the collision, he was totally unaware of her involvement.

When the master arrived on the bridge, he quickly tried to establish whether *Cosco Hong Kong* had collided with another vessel. He was aware that the vibration felt by the OOW and the chief officer must have been quite serious to cause the OOW to put the engine telegraph to 'stop'. However, several other factors also influenced the master's decision-making, including:

- The OOW was unsure what had happened but maintained that the vessel had been turning into clear water.

- There was no sign of a vessel in distress or any flotsam to indicate that a vessel had sunk.
- There was apparently no damage to *Cosco Hong Kong*.
- There were no distress alerts or messages on the GMDSS or AIS.
- Other than the communication from *SITC Pyeongtaek*'s OOW, there was no reaction by other vessels to the master's VHF broadcasts.
- There was no unusual activity among the fishing vessels in the area.

The master took over 1 hour to decide that *Cosco Hong Kong* should resume her passage. It is clear that he did not rush this decision. It is also clear from the master's concern that, had he been aware of a vessel in distress or sinking, he would have not hesitated to render assistance as required by UNCLOS (Paragraph 1.9.2), unlike the master of *Alam Pintar* (Paragraph 1.11.3). Although the provision of 'safety bonuses' similar to those offered by COSCO has been found to have had an adverse effect on the reporting of accidents, notably in the offshore sector of the shipping industry, this did not affect the master's decision-making on this occasion.

Nevertheless, in view of the uncertainty of *Cosco Hong Kong*'s involvement in a collision with another vessel or object, it would have been appropriate for the master to have manoeuvred his vessel towards the position where the impact had been felt. It would also have been prudent for the master to ask *SITC Pyeongtaek*'s OOW why he thought *Cosco Hong Kong* might have been in collision with a fishing vessel, to inform the MRCC Taizhou of the possible collision, and to save the vessel's SVDR data. Had these actions been taken, the search and rescue operation might have been triggered much sooner and the record of the OOW's actions would have been comprehensive.

The changes in the readings taken from the vessel's cathodic protection system, which were shown to the master during the morning of 6 March 2011, indicated that *Cosco Hong Kong* had lost some paint from the forward part of her hull since sailing from Hong Kong. However, although this would probably have increased the master's suspicion that his vessel had hit 'something', the continued absence of a distress message or alert over 8 hours after the incident would have supported his view that *Cosco Hong Kong* had not collided with another vessel.

## 2.8 BRIDGE WATCHKEEPING STANDARDS

Notwithstanding the current difficulties experienced in recruiting and retaining seafarers in China, COSCO Shanghai and other COSCO subsidiaries still currently maintain their own pools of deck officers, which have been trained in-house and are employed on board their managed vessels. Despite this control over the provenance of the seafarers on board the company's vessels, the circumstances of this accident strongly indicate that the second officer was not competent to keep his bridge watch in the conditions experienced.

During the events leading to the collision between *Cosco Hong Kong* and *Zhe Ling Yu Yun 135*, *Cosco Hong Kong*'s OOW failed to comply with fundamental requirements of the COLREGS (**Annex F**), the master's night orders (**Annex A**), the vessel's onboard instructions (**Annexes D and E**) and MCA guidance (Paragraph 1.8). In particular, he:

- Did not reduce the speed of *Cosco Hong Kong* when encountering concentrations of fishing vessels.
- Released the lookout from the bridge.
- Caused *SITC Pyeongtaek*'s OOW to take avoiding action, despite *Cosco Hong Kong* being the overtaking vessel.
- Sounded one single blast lasting about 3 seconds on the ship's whistle, which was potentially confusing to the watchkeepers on board other vessels in close proximity.
- Altered the container ship's heading to port when *Cosco Hong Kong* was the stand on vessel in a crossing situation.
- Did not monitor the effectiveness of his action and therefore did not see that *Zhe Ling Yu Yun 135* had turned to starboard and continued to close with his vessel.

These were serious departures from regulation, guidance and best practice that bring the OOW's knowledge and application of the COLREGS, his judgment, and his overall competency into question. Although the second officer was qualified for his rank, and had completed three contracts as a third officer and one contract as a second officer, aspects of his bridge watchkeeping on board *Cosco Hong Kong* on the morning of 6 March 2011 fell well short of the standard required by STCW.

The second officer had been trained by COSCO Shanghai and had previously served on board vessels that were managed by COSCO Shanghai and Cosmar (*Cosco Antwerp*). His training and employment was similar to other deck officers working on board vessels also managed by COSCO subsidiaries. Although the second officer's apparent lack of competency on 6 March 2011 might have been an aberration, the previous accidents involving Cosmar-managed vessels (Paragraph 1.11.1) and the unsatisfactory results of the rule of the road tests subsequently conducted, indicate that there is an underlying problem with the continuation and refresher training provided to deck officers by COSCO Shanghai, and possibly other COSCO subsidiaries.

Ship managers frequently assume that a deck officer's STCW qualification automatically guarantees a level of competency. The collision between *Cosco Hong Kong* and *Zhe Ling Yu Yun 135* and the recent similar accidents identified in Paragraph 1.11.3, clearly show that this is not the case. An OOW's knowledge and skills are perishable, and therefore need to be continually monitored, refreshed and developed.

However, judging the competency of bridge watchkeepers is not straightforward, and relies heavily on masters setting and demanding appropriate standards. In this case, the failure of the second officer's performance assessments to identify his lack of competency with regard to the COLREGs, indicates that these were not robust and their accuracy could not be relied upon.

Furthermore, although internal and external ISM audits and other ship visits by superintendents are generally useful in identifying documentary or procedural deficiencies (Paragraphs 1.6.3 and 1.6.4), these audits invariably occur when vessels are alongside and their usefulness in identifying competency issues among bridge watchkeeping officers is extremely limited. Other checks such as navigation audits, conducted while the vessels are at sea, the random scrutiny of VDR data, and the periodic testing of OOWs in key areas such as application of the COLREGS, are measures which are more likely to be effective in this respect.

## **2.9 FISHING VESSEL SAFETY**

### **2.9.1 Speed of loss**

The wreck of *Zhe Ling Yu Yun 135* lies only 30m from the position where the vessel last transmitted on AIS, which was at 0218.07, seconds before the collision. The fish transporter's contact with the container ship's bulbous bow at a closing speed in excess of 20kts would undoubtedly have resulted in catastrophic damage to the fish transporter. Given the position of the wreck, together with the lack of any radar or visual evidence of her being afloat after the collision, and the absence of any calls for assistance on VHF radio, it can only be concluded that *Zhe Ling Yu Yun 135* sank almost immediately after being struck. Consequently, as many of her crew would probably have been asleep in the vessel's accommodation, only the crew who were in the wheelhouse would have had any chance of escaping.

### **2.9.2 Survivability**

It is of serious concern that, although *Zhe Ling Yu Yun 135* was required to carry two liferafts secured with HRUs, neither liferaft floated to the surface and inflated after the vessel sank. Consequently, if any of her crew did manage to escape from the vessel, their chances of survival were considerably reduced. In addition, as the inflated liferafts would have been visible, a potential means of alerting *Cosco Hong Kong* and other vessels in close proximity was lost.

Although it is possible that the vessel's liferafts did not float to the sea surface because their HRUs malfunctioned or were damaged in the collision, it is equally likely that the liferafts were prevented from floating from the vessel as it sank by lashings intended to prevent their loss overboard or movement in bad weather. The lashing of liferafts in this manner is a dangerous practice which has been found among many fishing vessel crews worldwide, including the UK. As the potential effects of such practice on the lives of fishermen are severe, it must be discouraged if liferafts are to be available when needed.

It is also of serious concern that *Zhe Ling Yu Yun 135* was not reported to be missing until 18 hours after her collision with *Cosco Hong Kong*. She was not working with any of the fishing vessels in close proximity when she sank, and it was not recognised that she was missing until she failed to meet with fishing vessels operating 150nm offshore during the late afternoon of 6 March 2011.

The carriage of EPIRBs on board all Chinese fishing vessels would be highly impractical and would inevitably burden MRCCs with large numbers of false alarms. Indeed, the carriage of EPIRBs on board many Chinese fishing vessels which usually operate in groups or pairs relatively close to the shore, is probably unnecessary. In this case, however, the carriage of an EPIRB on board *Zhe Ling Yu Yun 135*, a fish transporter which frequently operated alone and was essentially 'ocean-going', would have quickly alerted the Chinese MRCCs, and a search would have started almost immediately after her sinking. Although this might not have increased the chances of survival for *Zhe Ling Yu Yun 135*'s crew, the carriage of EPIRBs on board similar vessels could help save many lives in the future.

### 2.9.3 The use of AIS

**Figures 2 to 5** show all vessels transmitting on AIS in the area. The large number of AIS targets would have crowded the port X-band radar display on board *Cosco Hong Kong*, particularly as the radar targets of the fishing vessels would have also been shown. Therefore, without the effective management of the received AIS data through the use of an appropriate range scale, the preference of *Cosco Hong Kong*'s OOW not to use this information was understandable. However, although the OOW denied himself valuable information concerning the identity of *Zhe Ling Yu Yun 135*, the vessel's heading and speed would have been available on the S-band ARPA radar.

AIS is a very useful aid to OOWs in identifying vessels, but when large numbers of vessels fitted with AIS are operating in the same area, the value of interfacing AIS information on radar, ECS, or ECDIS displays is reduced considerably. In such circumstances, although a vessel might transmit AIS information, it should not be taken for granted that this information has been seen and taken into account by the watchkeepers on board other vessels.

## SECTION 3 - CONCLUSIONS

### 3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. The use by *Cosco Hong Kong*'s OOW of one blast of approximately 3 seconds on the ship's whistle was incorrect and could easily have been mistaken on board *Zhe Ling Yu Yun 135* for one short blast, indicating that *Cosco Hong Kong* was altering course to starboard. [2.3.1]
2. *Cosco Hong Kong*'s alteration to port to avoid *Zhe Ling Yu Yun 135* was contrary to the requirements of the COLREGS, and would not have been expected by the crew on board *Zhe Ling Yu Yun 135*. [2.3.2]
3. When *Zhe Ling Yu Yun 135*'s heading was altered over 120° to starboard, neither this bold alteration nor the fish transporter's resulting approach were seen by *Cosco Hong Kong*'s OOW. [2.4.1]
4. The release of the AB lookout from *Cosco Hong Kong*'s bridge about 30 minutes before the collision was potentially pivotal to this accident. In allowing the lookout to leave the bridge at a critical time, when the vessel was on a night passage through dense shipping in adverse weather conditions, the second officer appears not to have appreciated the value of the lookout or the dangers developing ahead. [2.5]
5. *Cosco Hong Kong* could have avoided most of the fishing vessels in the vicinity had her voyage been planned further to the east, similar to the tracks of other vessels which were also transiting the area. [2.6]
6. Although *Cosco Hong Kong* was expected to encounter concentrations of fishing vessels, no consideration appears to have been given to amending the vessel's track, reducing speed, or enhancing the manning on the bridge. [2.6]
7. Several of the lessons to be learned from *Cosco Hong Kong*'s grounding in 2009 were not taken into account on this occasion. If further similar accidents are to be avoided in the future, it is important that masters and deck officers are made aware of the need to continually assess the risks to a vessel's safe navigation during a passage and, where necessary, implement appropriate mitigating measures. [2.6]
8. Although *Cosco Hong Kong*'s master took positive steps to determine whether his vessel had been involved in a collision, he did not take several precautions which might have triggered the search for *Zhe Ling Yu Yun 135* much sooner. [2.7]
9. The second officer's performance as the bridge watchkeeping officer on board *Cosco Hong Kong* on the morning of 6 March 2011 fell well short of the standard required by STCW. [2.8]
10. The second officer's apparent lack of competency, the previous accidents involving Cosmar managed vessels and the subsequent unsatisfactory results of the rule of the road tests, indicate that there is an underlying problem with the continuation and refresher training provided to deck officers by COSCO Shanghai, and possibly other COSCO subsidiaries. [2.8]

11. The second officer's performance assessments did not identify his lack of competency with regard to the application of the COLREGS, and the usefulness of internal and external ISM audits and other ship visits by superintendents in identifying competency issues among bridge watchkeeping officers was extremely limited. [2.8]

### **3.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS**

1. *Zhe Ling Yu Yun 135* was required to carry two liferafts secured with HRUs, but neither liferaft floated to the surface and inflated after the vessel sank. Although it is possible that the vessel's liferafts did not float to the sea surface because their HRUs malfunctioned or were damaged in the collision, it is equally likely that the liferafts were prevented from floating from the vessel as it sank by lashings intended to prevent their loss overboard or movement in bad weather. [2.9.2]
2. *Zhe Ling Yu Yun* was not reported to be missing until 18 hours after her collision with *Cosco Hong Kong*. Had the fish transporter been carrying an EPIRB, a search could have started almost immediately after her sinking. [2.9.2]